

5.3 Memory Footprint

Dataset	# Tokens	XTR				
		Index Size (GiB)				
		BruteForce	FAISS	ScaNN	WARP _(b=2)	WARP _(b=4)
BeIR [20]	NFCORPUS	1.35M	0.65	0.06	0.18	0.06
	SciFact	1.87M	0.91	0.08	0.25	0.07
	SCIDOCS	6.27M	3.04	0.28	0.82	0.24
	Quora	9.12M	4.43	0.43	1.21	0.35
	FiQA-2018	10.23M	4.95	0.46	1.34	0.38
	Touché-2020	92.64M	45.01	4.28	12.22	3.40
LoTTE [19]	Lifestyle	23.71M	11.51	1.08	3.12	0.88
	Recreation	30.04M	14.59	1.38	3.96	1.11
	Writing	32.21M	15.64	1.48	4.25	1.19
	Technology	131.92M	64.12	6.13	17.44	4.83
	Science	442.15M	214.93	20.57	58.46	16.07
	Pooled	660.04M	320.88	30.74	87.30	23.88
Total	–	1.44B	700.66	66.98	190.52	52.48

Table 4: Comparison of index sizes for the datasets. Note that PLAID’s memory usage is effectively identical to WARP’s, only slightly larger.

WARP’s primary advantage lies in its significant reduction in latency, though its benefits extend beyond speed alone. By adopting a ColBERTv2/PLAID-like approach for compression, WARP’s advantage over XTR also translates into a reduction in index size. This decrease in memory requirements broadens deployment options, particularly in resource-constrained environments. Table 4 compares index sizes across all evaluated test datasets. WARP_(b=4) demonstrates a substantially smaller index size compared to the BruteForce and ScaNN variants, providing a 7.3x and 2x reduction in memory footprint, respectively. Although the indexes generated by the FAISS implementation are marginally smaller, this comes at the cost of substantially reduced quality and latency. Notably, WARP_(b=2) outperforms the FAISS implementation in terms of quality with an even smaller index size¹⁶.

6 Conclusion

We present WARP, a highly optimized engine for multi-vector retrieval built upon ColBERTv2/PLAID and the XTR framework. WARP addresses key inefficiencies in existing systems through three major innovations: (1) WARP_{SELECT} for dynamic missing similarity imputation; (2) implicit decompression during retrieval; and (3) a streamlined two-stage reduction using dedicated C++ kernels. Together, these optimizations culminate in substantial performance gains, including a 41x speedup over XTR on LoTTE Pooled – cutting latency from over 6 seconds to just 171ms in single-threaded execution – and a 3x reduction in latency compared to ColBERTv2/PLAID, all without compromising retrieval quality. WARP also scales efficiently with increased thread count and offers a significantly reduced memory footprint, making it well-suited for deployment in resource-constrained environments.

Future work may incorporate techniques such as SIMD and GPU acceleration, more lightweight query encoding, and end-to-end training with WARP to better align model optimization with its retrieval process.

¹⁶ Additionally, WARP reduces memory requirements compared to PLAID as it no longer requires storing a mapping from document ID to centroids/embeddings.

References

- [1] Thibault Formal, Stéphane Clinchant, Hervé Déjean, and Carlos Lassance. 2024. Splate: sparse late interaction retrieval. (2024). arXiv: 2404.13950 [cs. IR].
- [2] Luyu Gao, Zhuyun Dai, and Jamie Callan. 2021. Coil: revisit exact lexical match in information retrieval with contextualized inverted list. (2021). <https://arxiv.org/abs/2104.07186> arXiv: 2104.07186 [cs. IR].
- [3] Stanford Future Data Systems Research Group. 2024. colbert-ir/colbertv2.0. <https://huggingface.co/colbert-ir/colbertv2.0>. (2024).
- [4] Stanford Future Data Systems Research Group. 2024. ColBERTv2/PLAID (Code). <https://github.com/stanford-futuredata/ColBERT>. (2024).
- [5] Ruiqi Guo, Philip Sun, Erik Lindgren, Quan Geng, David Simcha, Felix Chern, and Sanjiv Kumar. 2020. Accelerating large-scale inference with anisotropic vector quantization. (2020). <https://arxiv.org/abs/1908.10396> arXiv: 1908.10396 [cs. LG].
- [6] Hervé Jégou, Matthijs Douze, and Cordelia Schmid. 2011. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33, 1, 117–128. doi: 10.1109/TPAMI.2010.57.
- [7] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, (Eds.), Association for Computational Linguistics, Online, (Nov. 2020), 6769–6781. doi: 10.18653/v1/2020.emnlp-main.550.
- [8] Omar Khattab and Matei Zaharia. 2020. Colbert: efficient and effective passage search via contextualized late interaction over BERT. *CoRR*, abs/2004.12832. <https://arxiv.org/abs/2004.12832> arXiv: 2004.12832.
- [9] Ron Kohavi, Alex Deng, Brian Frasca, Toby Walker, Ya Xu, and Nils Pohlmann. 2013. Online controlled experiments at large scale. In *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '13)*. Association for Computing Machinery, Chicago, Illinois, USA, 1168–1176. ISBN: 9781450321747. doi: 10.1145/2487575.2488217.
- [10] Jinhuk Lee, Zhuyun Dai, Sai Meher Karthik Duddu, Tao Lei, Iftekhar Naim, Ming-Wei Chang, and Vincent Y. Zhao. 2024. google/xtr-base-en. <https://huggingface.co/google/xtr-base-en>. (2024).
- [11] Jinhuk Lee, Zhuyun Dai, Sai Meher Karthik Duddu, Tao Lei, Iftekhar Naim, Ming-Wei Chang, and Vincent Y. Zhao. 2024. Rethinking the role of token retrieval in multi-vector retrieval. (2024). arXiv: 2304.01982 [cs. CL].
- [12] Jinhuk Lee, Zhuyun Dai, Sai Meher Karthik Duddu, Tao Lei, Iftekhar Naim, Ming-Wei Chang, and Vincent Y. Zhao. 2024. XTR: Rethinking the Role of Token Retrieval in Multi-Vector Retrieval (Code). <https://github.com/google-deepmind/xtr>. (2024).
- [13] Minghan Li, Sheng-Chieh Lin, Barlas Oguz, Asish Ghoshal, Jimmy Lin, Yashar Mehdad, Wen-tau Yih, and Xilun Chen. 2022. Citadel: conditional token interaction via dynamic lexical routing for efficient and effective multi-vector retrieval. (2022). <https://arxiv.org/abs/2211.10411> arXiv: 2211.10411 [cs. IR].
- [14] Sean MacAvaney and Nicola Tonello. 2024. A reproducibility study of plaid. *arXiv preprint arXiv:2404.14989*.
- [15] Franco Maria Nardini, Cosimo Rulli, and Rossano Venturini. 2024. Efficient multi-vector dense retrieval using bit vectors. (2024). arXiv: 2404.02805 [cs. IR].
- [16] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2023. Exploring the limits of transfer learning with a unified text-to-text transformer. (2023). <https://arxiv.org/abs/1910.10683> arXiv: 1910.10683 [cs. LG].
- [17] Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: bm25 and beyond. *Foundations and Trends® in Information Retrieval*, 3, 4, 333–389. doi: 10.1561/15000000019.
- [18] Keshav Santhanam, Omar Khattab, Christopher Potts, and Matei Zaharia. 2022. Plaid: an efficient engine for late interaction retrieval. (2022). arXiv: 2205.09707 [cs. IR].
- [19] Keshav Santhanam, Omar Khattab, Jon Saad-Falcon, Christopher Potts, and Matei Zaharia. 2021. Colbertv2: effective and efficient retrieval via lightweight late interaction. *CoRR*, abs/2112.01488. <https://arxiv.org/abs/2112.01488> arXiv: 2112.01488.
- [20] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. Beir: a heterogeneous benchmark for zero-shot evaluation of information retrieval models. (2021). <https://arxiv.org/abs/2104.08663> arXiv: 2104.08663 [cs. IR].
- [21] Lee Xiong, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul Bennett, Junaid Ahmed, and Arnold Overwijk. 2020. Approximate nearest neighbor negative contrastive learning for dense text retrieval. (2020). <https://arxiv.org/abs/2007.00808> arXiv: 2007.00808 [cs. IR].
- [22] Jingtao Zhan, Jiaxin Mao, Yiqun Liu, Jiafeng Guo, Min Zhang, and Shaoping Ma. 2021. Optimizing dense retrieval model training with hard negatives. (2021). <https://arxiv.org/abs/2104.08051> arXiv: 2104.08051 [cs. IR].